Deep Learning-Based Radio Map for MIMO-OFDM Downlink Precoding

Wei Wang, Bin Yang, Wei Zhang

***Abstract*—Radio map is an advanced technology that mitigates the reliance of multiple-input multiple-output (MIMO) beamforming on channel state information (CSI). In this paper, we introduce the concept of deep learning-based radio map, which is designed to be generated directly from the raw CSI data. In accordance with the conventional CSI acquisition mechanism of MIMO, we first introduce two baseline schemes of radio map, i.e., CSI prediction-based radio map and throughput prediction-based radio map. To fully leverage the powerful inference capability of deep neural networks, we further propose the end-to-end structure that outputs the beamforming vector directly from the location information. The rationale behind the proposed end-to-end structure is to design the neural network using a task-oriented approach, which is achieved by customizing the loss function that quantifies the communication quality. Numerical results show the superiority of the task-oriented design and confirm the potential of deep learning-based radio map in replacing CSI with location information.**

***Keywords*—radio map, deep learning, MIMO, task-oriented approach**

I. Introduction

The availability of channel state information (CSI) is of pivotal importance to multiple-input multiple-output (MIMO) technology[1]. In 5G era, MIMO technology is characterized by the large-scale antenna, and to exploit the benefit of which, CSI at the transmitter side (CSIT) is regarded as indispensable. However, the increased antenna number, along with the low-cost hardware structure, e.g., hybrid analog and digital structure[2-4], renders the acquisition of CSI demanding for radio resources. In addition, in frequency division duplexing (FDD) massive MIMO, CSI feedback costs extra radio resources. Therefore, reducing the overhead of CSI acquisition and CSI feedback has become the major challenge in the research of massive MIMO, and various artificial intelligence (AI)-based algorithms have been proposed to enable channel estimation with insufficient pilots and lightweight CSI feedback[5-7].

On the other hand, since the development phase of 5G, several disruptive architectural changes of radio access network (RAN) have been proposed to meet the flexibility requirement driven by the rise of new technologies such as heterogeneous networks (HetNets) and centralized baseband signal processing[8]. In Ref. [9], the deployment of base stations (BSs) with different powers in HetNets calls for the decoupled downlink and uplink networks, as the uplink and downlink in HetNets are highly asymmetric. In Ref. [10], against the backdrop of cloud computing, the cloud radio access network (C-RAN) architecture, which exploits the benefits of the new communication and computing technologies, is proposed to achieve reduced cost, better energy efficiency, and improved spectrum utilization. In Ref. [11], inspired by the simple yet efficient signaling mechanism of the brain neurons[12], a fully-decoupled RAN (FD-RAN) has been proposed to physically decouple the control BSs and data BSs, and the data BSs are further split into uplink BSs and downlink BSs. The fully-decoupled and flexible RAN architecture, as an integral part of the cloud-native cybertwin-based network[13-14], can profoundly facilitate elastic resource cooperation to improve spectrum utilization rate, reduce energy consumption and improve the quality of user experience. However, the decoupled uplink and downlink networks and the cloud-based centralized baseband signal processing are a hinderance to the acquisition of CSI. Specifically, as the uplink BS differs from the downlink BS, the channel reciprocity cannot be utilized as in time division duplexing (TDD) MIMO. On the other hand, the CSI feedback from the mobile user to the downlink BS will suffer from unbearable delays as the information has to flow from the uplink BS to the downlink BS across different layers[11].

Beamforming gain of the large-scale antenna array can be

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achieved by directional transmission, either along the line-of-sight (LoS) path or the dominant none-line-of-sight (NLoS) paths, and the high-precision CSIT is generally not essential unless the channel is rich-scattering, e.g., Rayleigh fading channel. In that sense, location information can be utilized to replace CSIT given that the performance degradation is acceptable[15-21]. Hence, we resort to the radio map technology for an alternative solution to MIMO beamforming to accommodate the next-generation RAN. In Ref. [17], radio map is defined as the geographical signal power spectrum density as a function of location, frequency and time. In Refs. [18-19], the radio map is used to assist unmanned aerial vehicle (UAV) path planning by predicting the outage probabilities of UAVs. In Ref. [20], the concept of radio map is extended by including more channel-related information, e.g., pathloss, time of arrival (ToA), phase, azimuth and zenith angle of departure (AoD) and angle of arrival (AoA). With more channel information, the radio map has the potential to facilitate or even obviate sophisticated real-time CSI acquisition, offering solutions to practical challenges brought by the drastically increased communication overhead. In Ref. [21], the extended radio map is applied to environment-aware joint active and passive beam selection for intelligent reflecting surface (IRS)-aided wireless communication. In Ref. [22], an environment-aware and training-free beam alignment technique is proposed based on the extended radio map. The radio map technology is inherently data-driven. However, the channel data is collected at only a limited number of discrete locations, which is insufficient to form a complete map containing all the locations in the area of interest. To address this issue, a number of methods have been proposed, which can be roughly divided into three categories, namely interpolation, model-based, and deep learning methods. The interpolation method is a simple yet effective way to estimate the intermediate value with the limited and discrete measurement data. In Ref. [17], the Kriging interpolation[23] is used to estimate the covariance of signal power spectrum density (PSD), which is the optimal unbiased linear interpolation achieving the smallest variance of estimation error. In Refs. [20-22], the K nearest neighbors algorithm (KNN) and the inverse distance weighting (IDW) algorithm are used for interpolation. Although the interpolation methods are simple and easy to implement, its shortcomings are obvious. All measurement data needs to be saved to perform interpolation, which places strict demands on device memory and incurs huge storage overhead. In addition, when the geographical distribution of measurement data is uneven, finding the nearest several neighbors requires distance calculation and sorting, which will cause a huge computational burden when the number of the measurement data is large. The model-based methods assume that the channel follows a deterministic signal propagation model, and based on which a functional expression of the location and the corresponding channel information can be established[17]. In Ref. [24], a segmented regression based parameter estimation approach is proposed to reconstruct a finely structured map by exploiting both segmentation and signal strength models. However, the actual channel does not perfectly fit the mathematical model, sometimes resulting in significant errors in practice. Different from the model-based methods, the deep learning methods treat the functional relationship between location and channel information as a black box. In Ref. [25], a deep neural network (DNN) is applied to estimate the pathloss function in urban environment, which is accurate and computationally efficient.

Deep learning opens up a new direction for radio map designs. However, the current deep learning-based methods simply establish the relationship between location and the extracted channel information, e.g., pathloss, which fails to fully exploit the power of deep learning. It is worth mentioning that DNN has significant advantages in feature extraction from data. Hence, the deep learning-based radio map has the potential to directly process the high-dimensional raw CSI, and make a more profound impact on the signal transmission. Furthermore, the evaluation metric of the radio map design should be more closely connected to the quality of signal transmission. Therefore, we propose to study the deep learning-based radio map design with the data set of location and its corresponding raw CSI. Without loss of generality, we choose MIMO-orthogonal frequency division multiplexing (OFDM) downlink transmission as the specific communication task, and we further adopt spectral efficiency of the downlink transmission as the performance metric to assess the radio map. On the basis of the conventional TDD MIMO and FDD MIMO principles, we derive the codebook-based structure and non-codebook-based structure as the benchmark frameworks of radio map. We further propose the end-to-end structure that outputs the beamforming vector directly from the location information. By customizing a loss function that quantifies the communication quality, we steer the update direction of the neural network towards maximizing the communication quality. Numerical results demonstrate that the task-oriented design successfully achieves 90% of the optimum performance in different scenarios, showcasing the potential of utilizing deep learning-based radio maps to replace CSI with location information in the next-generation FD-RAN.

Notations: Column vectors (matrices) are denoted by bold-face lower (upper) case letters, $\boldsymbol{x}(n)$ denotes the nth element in the vector $\boldsymbol{x}$, $(\cdot)^*$, $(\cdot)^{\mathrm{T}}$, and $(\cdot)^{\mathrm{H}}$ represent conjugate, transpose, and conjugate transpose operation, respectively, $\|\cdot\|_2$ denotes the L2 norm of a vector.

II. System Model

Consider a MIMO-OFDM downlink of the FD-RAN, which consists of a downlink BS equipped with N_{t} transmit

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antennas and a user (UE) equipped with N_{r} receive antennas. Through downlink precoding, the frequency-domain received signal at the UE end is represented as

$$\boldsymbol{y}_k = \boldsymbol{H}_k \boldsymbol{f}_k s_k + \boldsymbol{w}_k, \tag{1}$$

where $\boldsymbol{H}_k \in \mathbb{C}^{N_{\mathrm{r}} \times N_{\mathrm{t}}}$ is the single-tap channel response for the kth sub-carrier, $\boldsymbol{f}_k \in \mathbb{C}^{N_{\mathrm{t}} \times 1}$ is the precoding vector, $s_k \in \mathbb{C}$ is the modulated signal, and $\boldsymbol{w}_k \in \mathbb{C}^{N_{\mathrm{r}} \times 1}$ is the additive white Gaussian noise (AWGN).

According to Shannon-Hartley theorem[26-27], the spectral efficiency of the kth sub-carrier is given by

$$R_k = \frac{C_k}{B_k} = \log\left(1 + \frac{\|\boldsymbol{H}_k \boldsymbol{f}_k\|_2^2}{\sigma_n^2}\right), \tag{2}$$

where C_k is the channel capacity of the kth sub-carrier, and B_k is the bandwidth of the kth sub-carrier. To optimize the downlink performance, the BS should find the optimal precoding matrix $\boldsymbol{F} = [\boldsymbol{f}_1, \boldsymbol{f}_2, \cdots, \boldsymbol{f}_K]$ that yields the maximum spectral efficiency over the K sub-carriers, i.e.,

$$\boldsymbol{F}^* = \arg\max_{\boldsymbol{F}} \sum_{k=1}^{K} \frac{R_k}{K}, \tag{3}$$

subject to the per-sub-carrier power constraint $\boldsymbol{f}_k^{\mathrm{H}} \boldsymbol{f}_k = 1$, $\forall k \in \{1, \cdots, K\}$.

However, the acquisition of the real-time CSI in FD-RAN faces great challenges due to its fully-decoupled structure of uplink base station and downlink base station. The objective of the radio map design is to construct a function that effectively depicts the relationship between the UE location $\boldsymbol{p} = (x, y, z)^{\mathrm{T}}$ and the optimal precoding matrix $\boldsymbol{F}$. In this regard, the radio map is represented as

$$\mathscr{M}(\boldsymbol{p}) = \hat{\boldsymbol{F}}. \tag{4}$$

With the radio map, the UE sends its location $\boldsymbol{p}$ to the core network via the uplink BS, and the downlink BS is then able to perform beamforming according to the predicted $\hat{\boldsymbol{F}}$.

To derive the radio map $\mathscr{M}(\cdot)$, deep learning techniques are applied to train a DNN with a collection of L location-CSI data tuples $D = \{\langle \boldsymbol{p}_l, \boldsymbol{H}_l \rangle\}_{l=1}^{L}$, where $\boldsymbol{H}_l = [[\boldsymbol{H}_{l,1}], [\boldsymbol{H}_{l,2}], \cdots, [\boldsymbol{H}_{l,K}]]$ is the CSI tensor that consists of the CSI of K sub-carriers. In this paper, we will explore the possible frameworks of constructing deep learning-based radio map and evaluate their performance in different scenarios.

III. Two Baseline Schemes of Radio Map Designs

Recall that the conventional MIMO precoding technique can be categorized into codebook-based precoding and non-codebook-based precoding[28]. Codebook is a set of precoding matrices that transform the modulated signal to the spatial domain. For codebook-based precoding, the precoding matrix index (PMI) that corresponds to the selected codeword is reported to the BS for downlink transmission. Codebook-based precoding is primarily used in FDD. For non-codebook-based precoding, the precoding matrix is not selected from a predefined codebook but is derived by exploiting the uplink/downlink channel reciprocity. In this section, we introduce two baseline frameworks of designing the radio map, in which the DNN is used to replace the channel sounding and acquisition component of the conventional non-codebook-based and codebook-based MIMO.

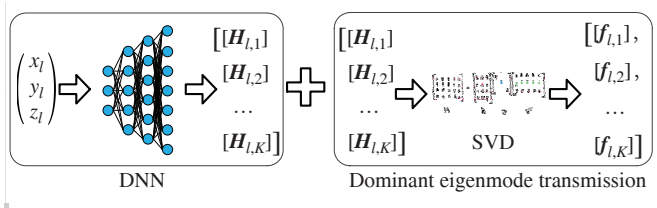

Fig. 1 The non-codebook-based structure with CSI prediction

A. Baseline Framework 1 – CSI Prediction-based Radio Map

In framework 1 (as shown in Fig. 1), the channel sounding and estimation process in the conventional pilot-based MIMO-OFDM downlink is replaced by a DNN $\mathscr{D}(\boldsymbol{p}, \boldsymbol{\theta}_a)$ with the parameter $\boldsymbol{\theta}_a$, which outputs the predicted CSI. Thus, the training data is $\langle \boldsymbol{p}_l, \boldsymbol{H}_l \rangle$, and the training policy is to minimize the expected error between the predicted output $\mathscr{D}(\boldsymbol{p}_l; \boldsymbol{\theta}_a) = \hat{\boldsymbol{H}}_l$ and the training data $\boldsymbol{H}_l$, i.e.,

$$\boldsymbol{\theta}_a^* = \arg\min_{\boldsymbol{\theta}_a} \frac{1}{L} \sum_{l=1}^{L} \mathrm{Loss}\left(\mathscr{D}(\boldsymbol{p}_l; \boldsymbol{\theta}_a), \boldsymbol{H}_l\right), \tag{5}$$

where l is the training sample index, and $\mathrm{Loss}(\cdot)$ is the minimum squared error (MSE) loss function.

With the predicted CSI, i.e., $\mathscr{D}(\boldsymbol{p}_l; \boldsymbol{\theta}_a) = \hat{\boldsymbol{H}}_l$, the $\hat{\boldsymbol{F}}_l$ can be derived for the optimal dominant eigenmode transmission (DET) that achieves the maximum received power[29]. Specifically, applying singular value decomposition (SVD), the channel response of the kth sub-carrier is decomposed as

$$\hat{\boldsymbol{H}}_{l,k} = \boldsymbol{V}_{l,k} \boldsymbol{\Sigma}_{l,k} \boldsymbol{U}_{l,k}, \tag{6}$$

where $\boldsymbol{\Sigma}_{l,k} = \mathrm{diag}\{\sigma_{l,k,1}, \sigma_{l,k,2}, \cdots, \sigma_{l,k,\mathrm{rank}(\boldsymbol{H}_k)}\}$, $\boldsymbol{U}_{l,k} = [\boldsymbol{u}_{l,k,1}, \boldsymbol{u}_{l,k,2}, \cdots, \boldsymbol{u}_{l,k,N_{\mathrm{t}}}]$ and the optimal beamforming vector is given by

$$\hat{\boldsymbol{f}}_{l,k} = \boldsymbol{u}_{l,k,1}, \tag{7}$$

which is the right singular vector corresponding to the maximum singular value $\sigma_{l,k,1}$.

We denote the mapping from $\boldsymbol{H} = [[\boldsymbol{H}_1] \cdots [\boldsymbol{H}_K]]$ to $\boldsymbol{F} = [\boldsymbol{f}_1, \cdots, \boldsymbol{f}_K]$ based on DET by $\mathscr{T}(\cdot)$. Then, the radio map for the non-codebook-based precoding is represented by

$$\mathscr{M}(\boldsymbol{p}; \boldsymbol{\theta}_a) = \mathscr{T}(\mathscr{D}(\boldsymbol{p}; \boldsymbol{\theta}_a)) = \hat{\boldsymbol{F}}. \tag{8}$$

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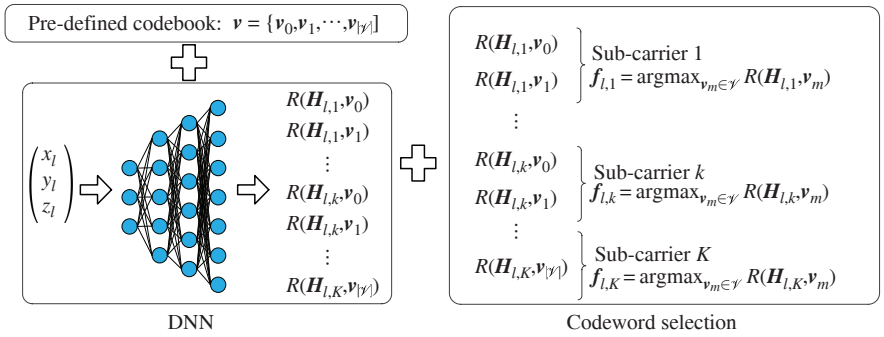

Fig. 2 The codebook-based structure with codeword-wise throughput prediction

Unlike the wireless applications of deep learning that use the discriminative models to map a high-dimensional input to the low-dimensional class labels/parameters, e.g., CSI fingerprinting localization[30], the network $\mathscr{D}(\boldsymbol{p};\boldsymbol{\theta}_a)$ is a mapping from the low-dimensional location to the high-dimensional CSI space. The network $\mathscr{D}(\boldsymbol{p};\boldsymbol{\theta}_a)$ interpolates the radio map and provides a means of predicting CSI at unmeasured locations.

B. Baseline Framework 2 – Throughput Prediction-based Radio Map

Codebook-based precoding was introduced in 4G-LTE and 5G-NR where predefined matrices, a.k.a., codewords, are evaluated and assigned to the transmitting antennas in the BS side. The selection of the optimal precoding matrix is performed in two ways. In the first way, the UE calculates the performance of codewords and reports back to the base station with the optimal codeword index[31-32]. In the other way, the UE quantizes its estimated CSI and feeds back the quantized CSI to the base station for the codeword selection in BS side[7].

In framework 2 (as shown in Fig. 2), the codeword evaluation of the predefined codebook is realized through a DNN $\mathscr{D}(\boldsymbol{p},\boldsymbol{\theta}_b)$ with the parameter $\boldsymbol{\theta}_b$. Analogous to the state-action table in Q-learning[33-34], the DNN is used to construct a table that outputs the predicted throughput of the candidate codeword $\boldsymbol{v}_m \in \mathscr{V}$ in the kth sub-carrier. Thus, the training data is $\langle \boldsymbol{p}_l, \boldsymbol{R}_l \rangle$, and the training policy is to minimize the expected error between the predicted output $\mathscr{D}(\boldsymbol{p}_l;\boldsymbol{\theta}_b) = \hat{\boldsymbol{R}}_l$ and the training data $\boldsymbol{R}_l$, i.e.,

$$\boldsymbol{\theta}_b^* = \arg\min_{\boldsymbol{\theta}_b} \frac{1}{L} \sum_{l=1}^{L} \text{Loss}\left(\mathscr{D}(\boldsymbol{p}_l;\boldsymbol{\theta}_b), \boldsymbol{R}_l\right), \tag{9}$$

where $\text{Loss}(\cdot)$ is the MSE loss function, and the (k,m)th element of the performance matrix $\boldsymbol{R}_l$ is

$$\boldsymbol{R}_l(k,m) = \log\left(1 + \frac{\|\boldsymbol{H}_{l,k}\boldsymbol{v}_m\|_2^2}{\sigma_n^2}\right). \tag{10}$$

With the predicted performance matrix, i.e., $\mathscr{D}(\boldsymbol{p}_l;\boldsymbol{\theta}_b) = \hat{\boldsymbol{R}}_l$, the optimal precoding matrix of the kth sub-carrier is determined by

$$\hat{\boldsymbol{f}}_{l,k} = \boldsymbol{v}_{m^*}, \tag{11}$$

where

$$m^* = \arg\max_{1 \leqslant m \leqslant |\mathscr{V}|} \hat{\boldsymbol{R}}_l(k,m). \tag{12}$$

We denote the codeword selection process by $\mathscr{S}(\cdot)$. Then, the radio map for the codebook-based precoding is represented by

$$\mathscr{M}(\boldsymbol{p};\boldsymbol{\theta}_b) = \mathscr{S}(\mathscr{D}(\boldsymbol{p};\boldsymbol{\theta}_b)) = \hat{\boldsymbol{F}}. \tag{13}$$

Remark 1 Radio map framework 1 and 2 are evolved from the conventional design paradigm of MIMO communications. Thus, the performance of framework 1 and 2 is inevitably restrained by the conventional design paradigm that is originally proposed for the non-data-driven system. For example, the codebook in framework 2 is independent of the CSI data and its mismatch with the CSI distribution might result in the degraded performance.

IV. Task-Oriented Designs of Radio Map

The radio map design is inherently data-driven. Thus, a brand new end-to-end structure that integrates multiple modules in the DNN has the potential to fully unleash the power of deep learning. In this section, we introduce the task-oriented designs of the end-to-end radio map.

A. The End-to-End Structure

Using the end-to-end structure (as shown in Fig. 3), the radio map is defined as

$$\mathscr{M}(\boldsymbol{p};\boldsymbol{\theta}_c) = \mathscr{D}(\boldsymbol{p};\boldsymbol{\theta}_c) = \hat{\boldsymbol{F}}, \tag{14}$$

where the input of the DNN model is $\boldsymbol{p}$, and the output of the DNN model is the precoding matrix $\hat{\boldsymbol{F}}$. The DNN model $\mathscr{D}(\boldsymbol{p};\boldsymbol{\theta}_c)$ is evaluated directly by the communication task.

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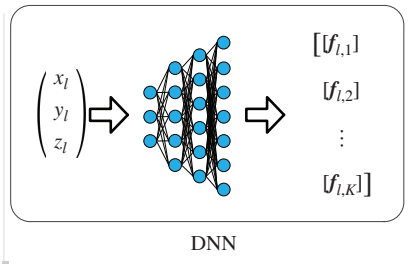

Fig. 3 The end-to-end structure

In other words, the DNN should maximize the spectral efficiency, i.e.,

$$\bar{\boldsymbol{R}} = \sum_{l=1}^{L}\sum_{k=1}^{K}\frac{1}{LK}\log\left(1+\frac{\|\boldsymbol{H}_{l,k}\hat{\boldsymbol{f}}_{l,k}\|_2^2}{\sigma_n^2}\right), \tag{15}$$

where $\hat{\boldsymbol{f}}_{l,k}$ is the precoding vector for the kth sub-carrier.

In the realm of machine learning, loss functions are usually different from the performance metrics of the task. For example, the widely-adopted loss function for the classification problems is cross entropy, while the performance metric is accuracy. Although increasing accuracy is of direct interest to human beings, the accuracy cannot be used as the loss function for training, as it is not differentiable. In our proposed end-to-end design, there are no off-the-shelves loss functions ready to be adopted. Thus, a suitable loss function is a critical factor for the success of the end-to-end structure.

B. Customizing Loss Function

To accommodate to the real-domain operations of deep neural networks, we introduce the following transformation from complex domain to real domain, i.e.,

$$\|\boldsymbol{H}_{l,k}\boldsymbol{f}_{l,k}\|_2^2 = \|\breve{\boldsymbol{H}}_{l,k}\breve{\boldsymbol{f}}_{l,k}\|_2^2, \tag{16}$$

where

$$\breve{\boldsymbol{H}}_{l,k} = \begin{pmatrix} \Re(\boldsymbol{H}_{l,k}) & -\Im(\boldsymbol{H}_{l,k}) \\ \Im(\boldsymbol{H}_{l,k}) & \Re(\boldsymbol{H}_{l,k}) \end{pmatrix}, \tag{17a}$$

$$\breve{\boldsymbol{f}}_{l,k} = \begin{pmatrix} \Re(\boldsymbol{f}_{l,k}) \\ \Im(\boldsymbol{f}_{l,k}) \end{pmatrix}. \tag{17b}$$

To align with the objective of maximizing spectral efficiency and, in the meantime, comply with the power constraint in the DNN, we customize the loss function as

$$\begin{aligned}&\mathrm{Loss}(\mathscr{D}(\boldsymbol{p}_l;\boldsymbol{\theta}_c),\breve{\boldsymbol{H}}_l) = \\ &-\sum_{k=1}^{K}\frac{1}{K}\log\left(1+\frac{\|\breve{\boldsymbol{H}}_{l,k}\bar{\mathscr{D}}_k(\boldsymbol{p}_l;\boldsymbol{\theta}_c)\|_2^2}{\sigma_n^2}\right),\end{aligned} \tag{18}$$

where $\bar{\mathscr{D}}_k(\boldsymbol{p}_l;\boldsymbol{\theta}_c) = \frac{\mathscr{D}_k(\boldsymbol{p}_l;\boldsymbol{\theta}_c)}{\|\mathscr{D}_k(\boldsymbol{p};\boldsymbol{\theta}_c)\|_2+\varepsilon}$ is the normalized precoding vector for the kth sub-carrier generated by the DNN model $\mathscr{D}(\boldsymbol{p}_l;\boldsymbol{\theta}_c)$, and ε is a very small positive number that is adopted to prevent the denominator from being 0, and, in the meantime, avoid causing significant bias of the vector norm. The training data of the end-to-end design is $\langle \boldsymbol{p}_l, \breve{\boldsymbol{H}}_l\rangle$, and the training policy is to minimize the expected loss function, i.e.,

$$\boldsymbol{\theta}_c^* = \arg\min_{\boldsymbol{\theta}_c}\frac{1}{L}\sum_{l=1}^{L}\mathrm{Loss}\big(\mathscr{D}(\boldsymbol{p}_l;\boldsymbol{\theta}_c),\breve{\boldsymbol{H}}_l\big). \tag{19}$$

To be a qualified loss function, (18) should be differentiable to enable gradient calculation and weights update.

Proposition 1 For a neural network $\mathscr{D}(\boldsymbol{p}_l;\boldsymbol{\theta}_c)$ with differentiable architecture, the customized loss function (18) is differentiable with respect to the parameter $\boldsymbol{\theta}_c$ on $\mathbb{R}^{N_{\boldsymbol{\theta}}}$.

Proof A differentiable function's derivative exists at each point in its domain. Thus, we need to prove that $\frac{\partial \mathrm{Loss}(\mathscr{D}(\boldsymbol{p}_l;\boldsymbol{\theta}_c),\breve{\boldsymbol{H}}_l)}{\partial\boldsymbol{\theta}_c}$ exists on $\mathbb{R}^{N_{\boldsymbol{\theta}}}$. Take the derivative of the loss function with respect to the parameters $\boldsymbol{\theta}_c$, we have

$$\begin{aligned}&\frac{\partial \mathrm{Loss}(\mathscr{D}(\boldsymbol{p}_l;\boldsymbol{\theta}_c),\breve{\boldsymbol{H}}_l)}{\partial\boldsymbol{\theta}_c} = \\ &-\sum_{k=1}^{K}\frac{1}{K}\frac{2\frac{\partial\bar{\mathscr{D}}_k^{\mathrm{T}}(\boldsymbol{p}_l;\boldsymbol{\theta}_c)}{\partial\boldsymbol{\theta}_c}\breve{\boldsymbol{H}}_{l,k}^{\mathrm{T}}\breve{\boldsymbol{H}}_{l,k}\bar{\mathscr{D}}_k(\boldsymbol{p}_l;\boldsymbol{\theta}_c)}{1+\frac{\|\breve{\boldsymbol{H}}_k\bar{\mathscr{D}}_k(\boldsymbol{p}_l;\boldsymbol{\theta}_c)\|_2^2}{\sigma_n^2}},\end{aligned} \tag{20}$$

where

$$\begin{aligned}&\frac{\partial\bar{\mathscr{D}}_k^{\mathrm{T}}(\boldsymbol{p}_l;\boldsymbol{\theta}_c)}{\partial\boldsymbol{\theta}_c} = \\ &\frac{\frac{\partial\mathscr{D}_k^{\mathrm{T}}(\boldsymbol{p}_l;\boldsymbol{\theta}_c)}{\partial\boldsymbol{\theta}_c}(\|\mathscr{D}_k(\boldsymbol{p}_l;\boldsymbol{\theta}_c)\|_2+\varepsilon)-\frac{\partial(\|\mathscr{D}_k(\boldsymbol{p}_l;\boldsymbol{\theta}_c)\|_2+\varepsilon)}{\partial\boldsymbol{\theta}_c}\mathscr{D}_k^{\mathrm{T}}(\boldsymbol{p}_l;\boldsymbol{\theta}_c)}{(\|\mathscr{D}_k(\boldsymbol{p}_l;\boldsymbol{\theta}_c)\|_2+\varepsilon)^2} = \\ &\frac{\frac{\partial\mathscr{D}_k^{\mathrm{T}}(\boldsymbol{p}_l;\boldsymbol{\theta}_c)}{\partial\boldsymbol{\theta}_c}((\|\mathscr{D}_k(\boldsymbol{p}_l;\boldsymbol{\theta}_c)\|_2+\varepsilon)\boldsymbol{I}-\frac{\mathscr{D}_k(\boldsymbol{p}_l;\boldsymbol{\theta}_c)\mathscr{D}_k^{\mathrm{T}}(\boldsymbol{p}_l;\boldsymbol{\theta}_c)}{\|\mathscr{D}_k(\boldsymbol{p}_l;\boldsymbol{\theta}_c)\|_2})}{(\|\mathscr{D}_k(\boldsymbol{p}_l;\boldsymbol{\theta}_c)\|_2+\varepsilon)^2}.\end{aligned} \tag{21}$$

As $\frac{\partial\mathscr{D}(\boldsymbol{p}_l;\boldsymbol{\theta}_c)}{\partial\boldsymbol{\theta}_c}$ exists and is bounded, $\frac{\partial\mathscr{D}_k^{\mathrm{T}}(\boldsymbol{p}_l;\boldsymbol{\theta}_c)}{\partial\boldsymbol{\theta}_c}$ in (21) exists and is bounded. In addition, since

$$\lim_{\|\mathscr{D}_k(\boldsymbol{p}_l;\boldsymbol{\theta}_c)\|_2\to 0}\frac{\mathscr{D}_k(\boldsymbol{p}_l;\boldsymbol{\theta}_c)\mathscr{D}_k^{\mathrm{T}}(\boldsymbol{p}_l;\boldsymbol{\theta}_c)}{\|\mathscr{D}_k(\boldsymbol{p}_l;\boldsymbol{\theta}_c)\|_2} = \boldsymbol{0}_{N_{\mathrm{t}}\times N_{\mathrm{t}}}, \tag{22}$$

$\frac{\partial\bar{\mathscr{D}}_k^{\mathrm{T}}(\boldsymbol{p}_l;\boldsymbol{\theta}_c)}{\partial\boldsymbol{\theta}_c}$ is bounded on $\mathbb{R}^{N_{\boldsymbol{\theta}}}$, including the region $\{\boldsymbol{\theta}|\boldsymbol{\theta}\in\mathbb{R}^{N_{\boldsymbol{\theta}}}, \|\mathscr{D}_k(\boldsymbol{p}_l;\boldsymbol{\theta})\|_2^2 = 0\}$, and thus $\frac{\partial\mathrm{Loss}(\mathscr{D}(\boldsymbol{p}_l;\boldsymbol{\theta}_c),\breve{\boldsymbol{H}}_l)}{\partial\boldsymbol{\theta}_c}$ exists on $\mathbb{R}^{N_{\boldsymbol{\theta}}}$.

Remark 2 When $\|\mathscr{D}_k(\boldsymbol{p}_l;\boldsymbol{\theta}_c)\|_2^2 = 0$, the loss function (18) achieves its maximized value, i.e., $\mathrm{Loss}(\mathscr{D}(\boldsymbol{p}_l;\boldsymbol{\theta}_c),\breve{\boldsymbol{H}}_l) = 0$. Thus, the update direction guided by the gradient of (18) is inherently biased from the region $\{\boldsymbol{\theta}|\boldsymbol{\theta}\in\mathbb{R}^{N_{\boldsymbol{\theta}}}, \|\mathscr{D}_k(\boldsymbol{p}_l;\boldsymbol{\theta})\|_2^2 = 0\}$. This, in turn, guarantees that $\frac{\mathscr{D}_k(\boldsymbol{p}_l;\boldsymbol{\theta}_c)}{\|\mathscr{D}_k(\boldsymbol{p};\boldsymbol{\theta}_c)\|_2+\varepsilon}\to\frac{\mathscr{D}_k(\boldsymbol{p}_l;\boldsymbol{\theta}_c)}{\|\mathscr{D}_k(\boldsymbol{p};\boldsymbol{\theta}_c)\|_2}$, and minimizes the gap between the customized loss function and the authentic performance metric. In addition, since the widely-adopted structures of neural network are

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Tab. 1 The role of DNN in the three schemes

Method	Output of the DNN	Loss function
Framework 1	CSI prediction: $\hat{\boldsymbol{H}}$	MSE
Framework 2	Codeword-wise throughput prediction: $\hat{\boldsymbol{R}}$	MSE
The end-to-end structure	Recommended precoding vector: $\hat{\boldsymbol{F}}$	Customized loss function

differentiable[35], the customized differentiable loss function, which aligns with the performance metric, allows for the effective optimization of the network's parameters through backpropagation.

C. Discussions

To differentiate between the two baseline schemes and the proposed task-oriented design, we have presented a comparison about the role of the DNN in Tab. 1. It can be seen that the DNNs in the baseline framework 1 and 2 play an intermediate role, which output the predicted CSI and the predicted codeword-wise throughput, respectively. The CSI prediction and throughput prediction are realized by training the DNN models to minimize the loss function of MSE. By contrast, the DNN in the end-to-end structure, which directly yields the precoding vector, is obtained by minimizing the customized loss function.

A notable distinction between applying the MSE loss function and the customized loss function described in equation (18) lies in the types of data they operate on. The MSE loss function takes inputs that share the same data type. Specifically, the two inputs are the CSI from the training data and the predicted CSI in the baseline framework 1, and the throughput computed from the training data and the predicted throughput in the baseline framework 2. By contrast, the customized loss function introduced in equation (18) involves two distinct data types as its inputs. The first input remains the CSI from the training data, while the second input is the precoding vector. In summary, a fundamental difference between the conventional MSE loss function and our customized loss function lies in the uniform data type of inputs in the former and the heterogeneous data types of inputs in the latter.

V. Numerical Results

To evaluate the performance of the aforementioned frameworks of radio map, we conduct numerical experiments using ResNet[36] as the DNN structure and Adam[37] as the parameter optimizing algorithm on the the DeepMIMO dataset[38] generated by Remcom Wireless InSite. The frequency $f_c = 3.5$ GHz and the bandwidth $B = 100$ MHz. The number of sub-carriers K is set to 1 024 and only the first 64 sub-carriers are considered. The transmit antennas and receive antennas

Tab. 2 The parameters of experimental scenarios

Parameter	Scenario 1	Scenario 2	Scenario 3
Name	O1	O1 Blockage	O2 dynamic
BS	1	3	1
The first row	1	501	301
The last row	1 000	1 500	400
Row sampling	0.2	0.2	1
User sampling	0.055 2	0.055 2	0.132 45
User number	2 000	2 000	2 000

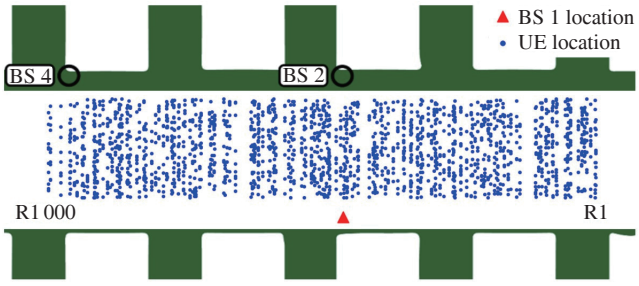

Fig. 4 The top view of scenario 1[38]

are both uniform linear array (ULA) and the number of transmit antennas $N_t = 32$ while the number of receive antennas $N_r = 2$. The transmit power is denoted by P. Moreover, we define the ratio of the number of training set samples to the total number of samples as training ratio α.

To validate the generalization performance of the proposed algorithm for different scenarios, we choose three outdoor scenarios of DeepMIMO. The first and the second scenario are basically the same. The only difference is that scenario 1 is unobstructed but scenario 2 is occluded. Therefore, in scenario 1, all channels have LoS paths while NLoS condition only exists in scenario 2. As for scenario 3, it is a dynamic scenario and its dynamism is reflected in the vehicles on the road constantly changing their positions with every scene. The movement of the vehicles will change the CSI from the BS to the UE. The scenes are continuously sampled with a sampling time of 100 ms, and ten scenes are selected in the simulation. The detailed parameters of the three chosen scenarios are listed in Tab. 2 and the top view of scenario 1 is shown in Fig. 4.

The spectrum efficiencies of the proposed frameworks and the theoretical optimums in these three scenarios are shown in Fig. 5. The transmit power $P = 0$ dBm and the training ratio $\alpha = 0.5$. In Fig. 5, all of these three frameworks can improve spectral efficiencies in different scenarios. The end-to-end framework is the best among these three frameworks and its spectral efficiency is very close to the optimum, which is approximately 90% of the optimum. The performance of the framework 2 is the middle of these three frameworks, and is better than the framework 1. The framework 2, i.e., the ra-

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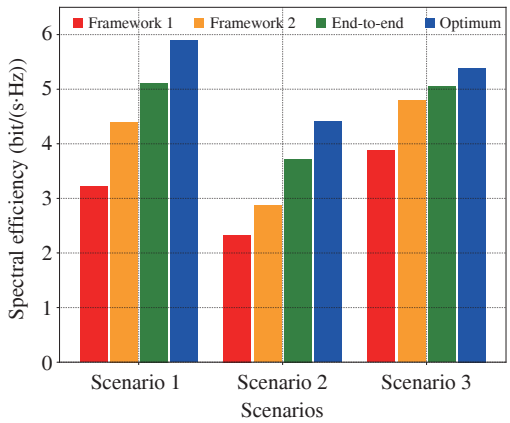

Fig. 5 The spectral efficiencies in three different scenarios

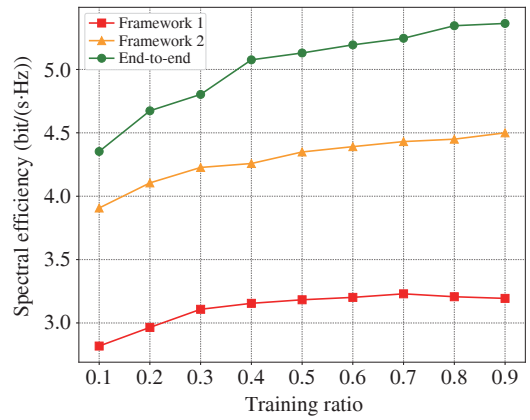

Fig. 6 The spectral efficiency vs. training ratio

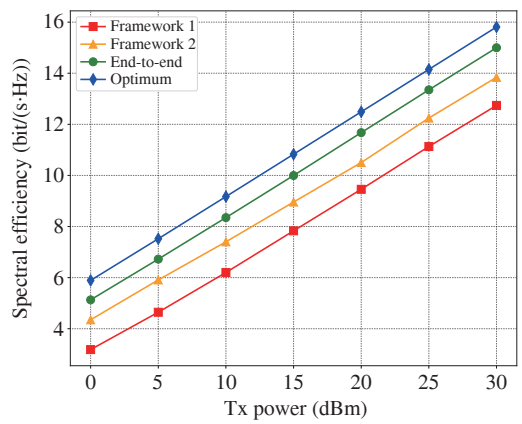

Fig. 7 The spectral efficiency vs. Tx power

dio map for codebook-based precoding scheme can achieve nearly 70% of the optimal result. Although the framework 1 is the worst, its performance still exceeds 50% of the optimum in all the scenarios. The results reveal that using end-to-end structure combined with task-oriented design can exert near optimal performance. As for framework 1 and framework 2, their performances are restrained by the conventional design paradigm, which prevents them from achieving better performance. Compared to the framework 2, in the framework 1, the task is only to learn how to select codewords in a given codebook, which is much easier than learning the entire channel matrix in the framework 2. Therefore, the performance of the framework 1 is better than the framework 2.

In Fig. 6, we show the spectral efficiencies of the proposed frameworks under different training ratios in scenario 1. The transmit power P is set to 0 dBm. In general, the spectral efficiencies of the three frameworks increase with the training ratio α. Specifically, the end-to-end framework increase significantly with α compared with the framework 1 and the framework 2. For the framework 1, its spectral efficiency remains basically unchanged when α is larger than 0.5. In this case, the continuous increase of training samples does not bring more improvements to the spectral efficiency. For the framework 2, its growth is relatively slow but it will still increase with α. These three frameworks can still work even under very low training ratio, i.e., $\alpha = 0.1$, which shows their potentials in the case of less training samples. In addition, the results also indicate that framework design is more important than training ratios. For example, the result of the end-to-end framework under $\alpha = 0.1$ and $\alpha = 0.2$ is better than all the results of the framework 1 and the framework 2, respectively.

Then, we investigate the effects of the transmit power on the spectral efficiencies of these three frameworks in scenario 1, as well as the optimum in Fig. 7. The training ratio α is set to 0.5. The spectral efficiencies of three frameworks and optimum increase almost linearly with P. To be specific, for every 5 dBm increase in transmit power, the spectral efficiency will increase by nearly 2 bit/(s·Hz) for all the frameworks. The results illustrate that by increasing the transmit power, the signal-to-noise ratio (SNR) can be effectively improved, as well as the spectral efficiency.

The training losses of the proposed frameworks vs. epoches are shown in Fig. 8, in which $\alpha = 0.5$, $P = 0$ dBm. As shown in Fig. 8, the training losses of three frameworks decrease monotonously with the number of epoches. The end-to-end framework converges fastest and only needs 100 epoches to achieve a good performance, which indicates that customizing the loss function can make deep learning more efficient. Different from the end-to-end framework, the framework 1 and the framework 2 apply MSE as loss function. It can be seen that the rates of convergence of both two frameworks are obviously inferior to the end-to-end framework.

VI. Conclusion

In this paper, we study the deep learning-based radio map technology and evaluate its capability in replacing CSI with location information for the next-generation FD-RAN. Three frameworks of radio map are proposed and evaluated using the DeepMIMO dataset. The end-to-end structure, enabled

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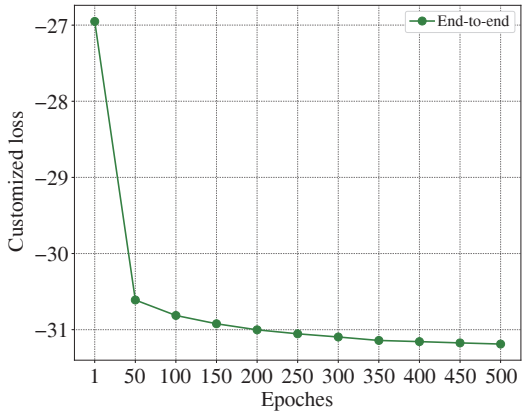
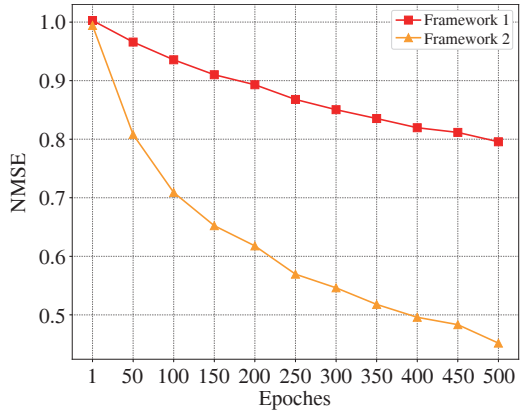


(a)



(b)

Fig. 8 The training loss vs. number of epoches: (a) The end-to-end structure with the customized loss function; (b) The baseline framework 1 and 2

by the task-oriented design with the customized loss function, achieves 90% of the optimum performance in different scenarios and is thus regarded as a promising design paradigm for deep learning-based radio map.

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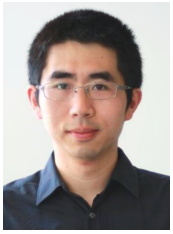

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